This assignment will not be graded and is only for practice.

Level 1

1) Suppose $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which of the following options are correct? ***1 point***

- [ ] The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- [ ] The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.
- [ ] T is neither one-one nor onto.
- [ ] T is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

T is neither one-one nor onto.

2) If the matrix corresponding to a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, with respect to standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain, is $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$. Which of the following is the appropriate definition of T? ***1 point***

- ( ) $T(x, y) = (2x + y, 3x + 4y)$
- ( ) $T(x, y) = (2x + y, 3x - 4y)$
- ( ) $T(x, y) = (x + 4y, 2x + 3y)$
- ( ) $T(x, y) = (2x + 3y, x + 4y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y) = (2x + 3y, x + 4y)$

Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of $\mathbb{R}^3$. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$ be a basis of W. Let $T : W \to \mathbb{R}^2$ be a linear transformation such that $T(2, 1, 0) = (1, 0)$ and $T(1, 0, 1) = (0, 1)$. Answer the questions 3, 4 and 5 using the given information.

3) Which of the following is the appropriate definition of T? **_1 point_**

- [ ] $T(x, y, z) = (x, y)$
- [ ] $T(x, y, z) = (x, z)$
- [ ] $T(x, y, z) = (y, z)$
- [ ] $T(x, y, z) = (x - y - z, x - 2y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y, z) = (y, z)$

$T(x, y, z) = (x - y - z, x - 2y)$

4) Choose the correct options. **_1 point_**

- ○ T is one to one but not onto.
- ○ T is onto but not one to one.
- ○ T is neither one to one nor onto.
- ○ T is an isomorphism.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is an isomorphism.

5) What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for $\mathbb{R}^2$? **_1 point_**

- $\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$
- $\frac{1}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$
- $\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$
- $\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$

6) Choose the set of correct options. ***1 point***

- [ ] Let $u = (3, 1, 0)$, $v = (0, 1, 7)$, and $w = (3, 0, -7)$. There is a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ such that $T(u) = T(v) = (0, 0, 0)$ and that $T(w) = (5, 1, 0)$.
- [ ] Let $u = (2, 1, 0)$, $v = (1, 0, 1)$, and $w = (3, 5, 6)$. There is a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^2$ such that $T(u) = (1, 0)$, $T(v) = (0, 1)$ and that $T(w) = (5, 6)$.
- [ ] Let $u = (1, 0, 0)$, $v = (1, 0, 1)$, and $w = (0, 1, 0)$. There is a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ such that $T(u) = (0, 0, 0)$, $T(v) = (0, 0, 0)$, $T(w) = (0, 0, 0)$, and $T(1, 4, 2) = (2, 4, 1)$.
- [ ] Let $u = (1, 0)$, and $v = (0, 1)$,. There is a linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ such that $T(u) = (\pi, 1)$, and $T(v) = (1, e)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $u = (2, 1, 0)$, $v = (1, 0, 1)$, and $w = (3, 5, 6)$. There is a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^2$ such that $T(u) = (1, 0)$, $T(v) = (0, 1)$ and that $T(w) = (5, 6)$.
Let $u = (1, 0)$, and $v = (0, 1)$,. There is a linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ such that $T(u) = (\pi, 1)$, and $T(v) = (1, e)$.

Level 2

7) Choose the set of correct options. ***1 point***

- [ ] If the Identity matrix of order 2 is the matrix representation of a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of T with respect to any other basis β for both the domain and co-domain.
- [ ] Let T be an isomorphism between two vector spaces, and A be the matrix representation of T with respect to some basis. Then the nullity of A is 0.
- [ ] If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T : \mathbb{R}^2 \to \mathbb{R}^2$ with respect to the standard ordered basis of $\mathbb{R}^2$, for both the domain and co-domain, then the matrix representation of $T \circ T$ with respect to the same bases will also be A.
- [ ] If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T : \mathbb{R}^2 \to \mathbb{R}^2$ with respect to the standard ordered basis of $\mathbb{R}^2$, for both the domain and co-domain, then the matrix representation of $T \circ T$ will be the identity matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of T with respect to any other basis β for both the domain and co-domain.

Let T be an isomorphism between two vector spaces, and A be the matrix representation of T with respect to some basis. Then the nullity of A is 0.

If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T : \mathbb{R}^2 \to \mathbb{R}^2$ with respect to the standard ordered basis of $\mathbb{R}^2$, for both the domain and co-domain, then the matrix representation of $T \circ T$ with respect to the same bases will also be A.

Let $\beta = \{(1,0),(0,1)\}$ and $\gamma = \{(1,1),(1,-1)\}$ be two bases of $\mathbb{R}^2$. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ with respect to β for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

8) What is the matrix representation of T with respect to γ for the domain and β for the co-domain? ***1 point***

- $\begin{bmatrix} a+b & c+d \\ a-b & c-d \end{bmatrix}$
- $\begin{bmatrix} a & -b \\ c & -d \end{bmatrix}$
- $\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$
- $\begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

9) What is the matrix representation of T with respect to γ for both the domain and the co-domain? ***1 point***

- $\begin{bmatrix} a+b+c+d & a-b+c-d \\ a+b-c-d & a-b-c+d \end{bmatrix}$
- $\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a-b+c-d}{2} \\ \frac{a+b-c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$
- $\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a+b-c-d}{2} \\ \frac{a-b+c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$
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$$\begin{bmatrix} a+b+c+d & a+b-c-d \\ a-b+c-d & a-b-c+d \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a-b+c-d}{2} \\ \frac{a+b-c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$$

10) Which of the following options are correct? ***1 point***

- [ ] If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation which is both one-one and onto then T^{-1} is also a linear transformation.
- [ ] If $T : \mathbb{R}^n \to \mathbb{R}^m$ then the matrix corresponding to T, with respect to the standard bases for R^n and R^m is of order $n \times m$.
- [ ] If $T : \mathbb{R}^n \to \mathbb{R}^m$ then the matrix corresponding to T, with respect to the standard bases for $\mathbb{R}^n$ and $\mathbb{R}^m$ is of order $m \times n$.
- [ ] Consider the vector space $V = \{\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{R}, \}$, that is, the set of all 2×2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from $\mathbb{R}^3$ to V.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation which is both one-one and onto then T^{-1} is also a linear transformation.

If $T : \mathbb{R}^n \to \mathbb{R}^m$ then the matrix corresponding to T, with respect to the standard bases for $\mathbb{R}^n$ and $\mathbb{R}^m$ is of order $m \times n$.

Consider the vector space $V = \{\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{R}, \}$, that is, the set of all 2×2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from $\mathbb{R}^3$ to V.

Your score is: 0/10